Alex meeting — how we fit λ, γ, and temperature (plain English)

For: Charles — read aloud or skim before / with Alex
Date: 2026-08-20
Slides: `3-Master_Plan/re_entry/HEROs_and_PASSes/slides/auto/CHAR_PD21_MLE_fit_AUTO.pptx`
Script: `sports/scripts/pd21_draft_bernoulli_mle.py`
Deep dive: `MLE_basics.md`, `PD20_softmax_K_winners_explainer.md`

One sentence for Alex

We used **maximum likelihood** on real NCAA draft outcomes to fit **how much congestion (λ) and ability scaling (t)** enter the draft-probability model, with **γ** fixed at our sim default; a separate **temperature sweep** showed the generative inverted-U survives soft SELECT before we ran that fit.

Start here: three steps, not one blob

The basketball pipeline is deliberately **three steps**:

1. **ASSIGN** — who is on which roster (homophily **ρ**).
2. **SCORE** — rank players: ability minus congestion (**λ**, **γ**).
3. **SELECT** — who gets drafted: softmax with **temperature t**.

Binding rule: score ≠ select. Fitting draft outcomes tells you about the **draft probability model** (SELECT + how score enters it). It does **not** automatically fit ρ. ρ is calibrated separately against roster-sorting statistics (H_sort bracket search — not draft MLE).

For today's meeting, focus on **SCORE + SELECT** parameters: **λ, γ, t**.

What "MLE" means here (elementary)

Maximum likelihood = pick parameters that make the **observed draft data as plausible as possible** under a written-down model.

For each college player-season we observe **Y = 1** if that player was ever drafted, **Y = 0** if not.

We assign a model probability **p_i** that player *i* gets drafted. Then we ask: if these parameters were true, how likely is the **whole panel** of 0s and 1s we actually saw?

We multiply (or sum logs) over all players and **adjust λ, t, (γ)** until that probability is as large as possible. That's MLE.

No new experiment — one historical panel (2013–2021 for the committed fit: **38,123** player-seasons, **882** drafted, POST-QC panel, min 20 minutes).

The model we fit (Alex PD21 lock)

Draft probability for player *i* in a season (softmax within that season):

$$p_i \propto \exp(A_i/t)\,\exp(-\lambda L_i^C)$$

Plain English:

- **A_i** — ability (PPM z-score within season).
- **L^C_i** — LOO congestion / pool pressure on the roster.
- **t** — temperature: smaller t → draft odds more concentrated on top ability; larger t → flatter.
- **λ** — how hard congestion penalizes draft odds in the **score**.
- **γ** — sharpness of the viability map that builds **L^C** from rosters (enters through congestion, not as a free draft label).

Likelihood (Bernoulli):

$$\ell = \sum_i \left[Y_i \log p_i + (1 - Y_i)\log(1 - p_i)\right]$$

Critical Alex point: K (number of draft slots) is not in this formula. Probabilities sum to 1 within each season by construction. We fit **who** gets high draft probability, not "make ∑p_i = K."

That is **standard MLE** for independent Bernoulli outcomes with a parametric **p_i(θ)** — just not the older "multinomial exactly K winners" story.

Why this instead of "standard MLE with K draws"?

Before PD21 we worried: the **simulator** picks **K** draftees with weighted draws without replacement. Should the likelihood match that exactly?

Alex's answer (Aug 14): For v1, **Bernoulli + softmax is OK.**

Approach	Idea	Our status
Bernoulli MLE (what we ran)	Each player draft / not-draft; **p_i** from softmax	**Committed fit**
K-draw likelihood	Likelihood matches sim's without-replacement draws	Hero **sim** may keep this; not required for empirical MLE
Multinomial "exactly K picks"	∑p_i constrained to K	**Not** Alex v1

So when you say to Alex: *"Yes — I used MLE on the empirical draft data"* — that is accurate for **λ** and **t** (and optionally **γ**) under the **Bernoulli** model he approved.
The sim can still use K-draws for generative pictures; Alex expects only a small gap at MBB scale.

How we actually optimized (not hand-waving)

1. **Build panel** — real rosters, ability, congestion, draft labels (Y_draft).
2. **Coarse grid** — try many (λ, t) pairs (e.g. λ ∈ {0.5,...,3}, t ∈ {0.01,...,10}) to get a good starting point.
3. **L-BFGS-B** — smooth numerical optimizer that **maximizes log-likelihood** (minimizes negative log-lik).
4. **γ profile** — fix γ at several values (5, 8, 10, ..., 200); at each γ, re-run grid + BFGS for (λ, t) and plot max ℓ vs γ.

Script: `sports/scripts/pd21_draft_bernoulli_mle.py`
Artifacts: `3-Master_Plan/re_entry/HEROs_and_PASSes/pd21_mle/`

What we found (committed numbers, 2013–2021)

Parameter	Estimate	Plain read
$\hat{\lambda}$	**≈ 2.57**	Congestion in the score **matters** for matching draft outcomes (λ > 0).
$\hat{t}$	**≈ 1.07**	Moderate softmax — not ice-cold (t→0) and not uniform (t→∞).
γ	**18 fixed**	Tier-1 sim default; profile shows ℓ **flat** from γ ≈ 18–100, so γ is **not sharply identified** by draft labels alone.
Max ℓ	**≈ −6919.1**	Best fit under fixed γ = 18.

γ profile takeaway: Best log-lik on the grid is near **γ = 100**, but **$\hat{\lambda}$ and $\hat{t}$ barely move** (≈ 2.50–2.57 and ≈ 1.06–1.07). We report **γ = 18** to stay aligned with the generative sim, not because draft MLE pins γ tightly.

Sanity checks (not the headline):

- Mean "top-K recall" on drafted players ≈ **12%** — soft ranking partially overlaps real draftees (draft is sparse).
- Mean overlap with empirical top-K per season ≈ **12** picks — reasonable for a probability model, not a deterministic ranker.

Where "temperature" appears twice (don't conflate)

A. PD20 generative sweep (before MLE)

Purpose: Prove the **inverted-U** in sim draft rate vs teammate quality **does not break** when SELECT is Gibbs/soft instead of hard top-K.

Method: Grid over **$\log_{10}(t)$** with fixed λ panels {1.5, 2.0} on the **sim bridge** (rule D). **Not** MLE on Y_draft.

Finding: Inverted-U-like curvature **persists** across most of the temperature grid → **gate cleared** for fitting.

Plot: `pd20_temperature/GRANDCHILD_temperature_select_sweep_2013_2021.png`

B. Empirical Bernoulli MLE (the fit Alex cares about)

Purpose: Fit **t** (and λ) so model draft probabilities **best explain real** draft/not-draft outcomes.

Finding: $\hat{t}$ ≈ 1.07 — a concrete empirical temperature on the same 2013–2021 window.

When Alex asks "what's t?" — give **$\hat{t}$ from MLE**. Mention PD20 sweep only as **why we were allowed to move to MLE** after soft SELECT.

What about ρ and γ?

Knob	Layer	How fit	MLE?
ρ	ASSIGN	Bracket search on H_sort vs sim	**No** — calibration, not draft likelihood
λ, t	SCORE + SELECT	Bernoulli MLE on Y_draft	**Yes**
γ	SCORE (via L^C)	Fixed at 18 for committed run; profile shows flat ℓ	**Partially** — not sharply identified

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What to say about the hero (Layer A) — parked for later

If Alex asks about the **empirical draft-rate vs teammate-quality curve** (the "hero"):

- On **cleaned** data (team-seasons with ≥10 games, min 20 minutes): **middle rise** in draft rate with better teammates; **elite tail is flat**, not a sharp inverted-U dip.
- The **July inverted-U tail** replays only on **pre-QC** data (cameo contamination in the top bin — documented in `pass_a/sensitivity/`).
- **Generative Pass B/C** still show inverted-U in sim — that is the mechanism story, not bin-for-bin match to ventiles.

One line: *"Rung 1 on real data is closed with honest limits; today I'm showing the* ***fit*** *of λ, t, γ on draft outcomes. Hero shape is a separate QC thread we can walk through if you want."*

Do **not** lead with hero angst unless he asks — lead with **MLE fit + generative knockout**.

Suggested talk track (5 minutes)

1. **Three layers** — ASSIGN / SCORE / SELECT; today is draft MLE on the last two.
2. **PD20 slide** — soft SELECT doesn't kill inverted-U in sim (gate).
3. **Bernoulli MLE** — formula, K not in likelihood, L-BFGS-B on real Y_draft.
4. **Numbers** — $\hat{\lambda} \approx 2.57$, $\hat{t} \approx 1.07$, $\gamma = 18$ (profile flat).
5. **γ profile slide** — congestion matters; γ not tight from draft alone.
6. **If time:** hero is middle-rise + flat tail on POST-QC; July dip was sensitivity only.

If Alex pushes back

"Is this really MLE?"
Yes — we maximize the Bernoulli log-likelihood over (λ, t) with standard numerical optimization. See JSON: `pd21_mle/PD21_draft_bernoulli_mle_2013_2021.json`.

"Why fix γ?"
Draft labels don't pin γ sharply; we anchor to sim (18) for a single deliverable. Joint 3-param MLE is in the script for the next pass.

"What about K?"
Alex lock: K not in the formula; probabilities sum to 1 per season. Sim may still use K-draws for pictures.

"ρ?"
Separate ASSIGN calibration ($\rho^* \approx 0$ on locked panel). Draft MLE uses **fixed empirical rosters**.

Files to have open

Item	Path
Slide deck	`slides/auto/CHAR_PD21_MLE_fit_AUTO.pptx`
MLE JSON	`pd21_mle/PD21_draft_bernoulli_mle_2013_2021.json`
Temperature sweep	`pd20_temperature/GRANDCHILD_temperature_select_sweep_2013_2021.png`
γ profile	`pd21_mle/PD21_draft_bernoulli_mle_2013_2021_gamma_profile.png`
Regenerate slides	`python sports/scripts/build_pd21_mle_fit_slides.py --slides-only`

Closing line

"We cleared soft SELECT, then ran Bernoulli MLE on real draft outcomes: $\hat{\lambda} \approx 2.6$, $\hat{t} \approx 1.1$, γ anchored at 18. Congestion in the score is needed to match the data. The empirical hero ventile is a separate, documented read — middle rise on cleaned data, flat elite tail — and the generative model still does the inverted-U work in sim."